

Thrust Areas for Airborne Surveillance Antenna Structures

Centre for Airborne Systems (CABS) is a nodal agency of the Defence Research and Development Organization (DRDO) for Design, Development and evaluation of airborne surveillance system for military application. CABS is mandated to develop the Airborne Early Warning & Control (AEW&C) System for IAF and other Defence services. Towards this, CABS has developed core competencies in the field of structural design and development of Airborne Surveillance System components and assemblies. Integration and testing of the surveillance antenna structure to meet the airworthiness certification requirements, the following thrust areas in which CABS Structural Design and analysis team works.

- 1) Design of Radome Structure for the Airborne Surveillance Antenna to meet Aerodynamic, Structural & Electromagnetic functional requirements.
- 2) Finite Element Structural analysis to meet FAR 25 / Equivalent strength requirements.
- 3) Ground Vibration Testing (GVT) as per MIL standards
- 4) Static Strength Testing (SST) and Modal Analysis testing (MAT) – FAR 25 FAR 25 / Equivalent requirements.
- 5) Lightning testing on the Antenna Structure as per RTCA-DO-160.
- 6) Coupon testing of Composite Material – MIL HDBK 17.
- 7) Bird strike – Explicit Finite Element & testing to meet FAR 25 FAR 25 / Equivalent requirements.
- 8) Structural Health Monitoring (SHM) for composite structures.